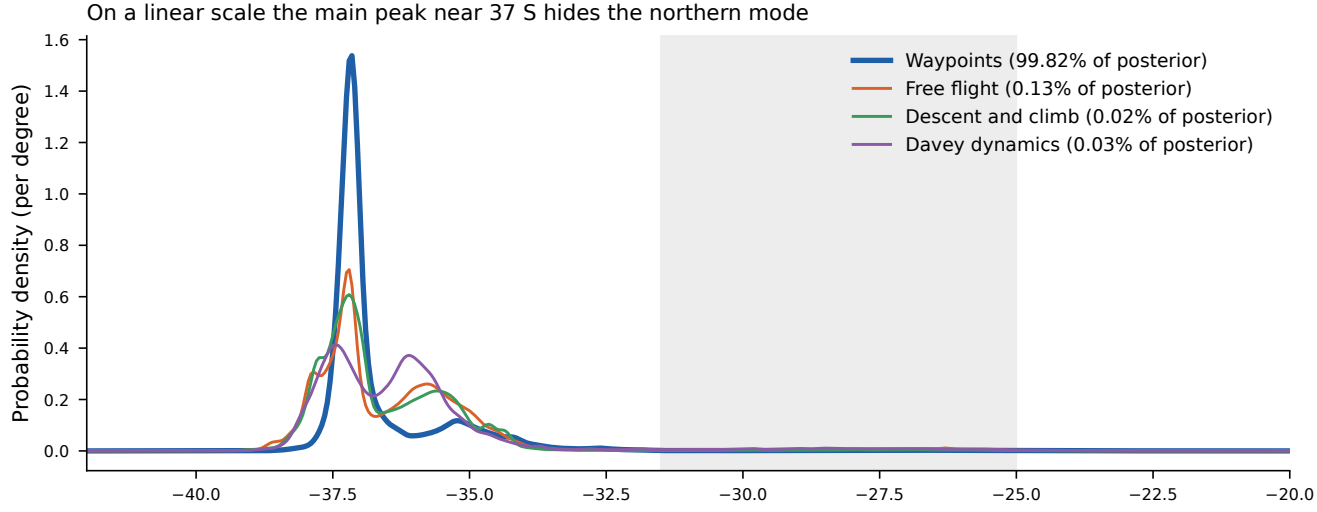
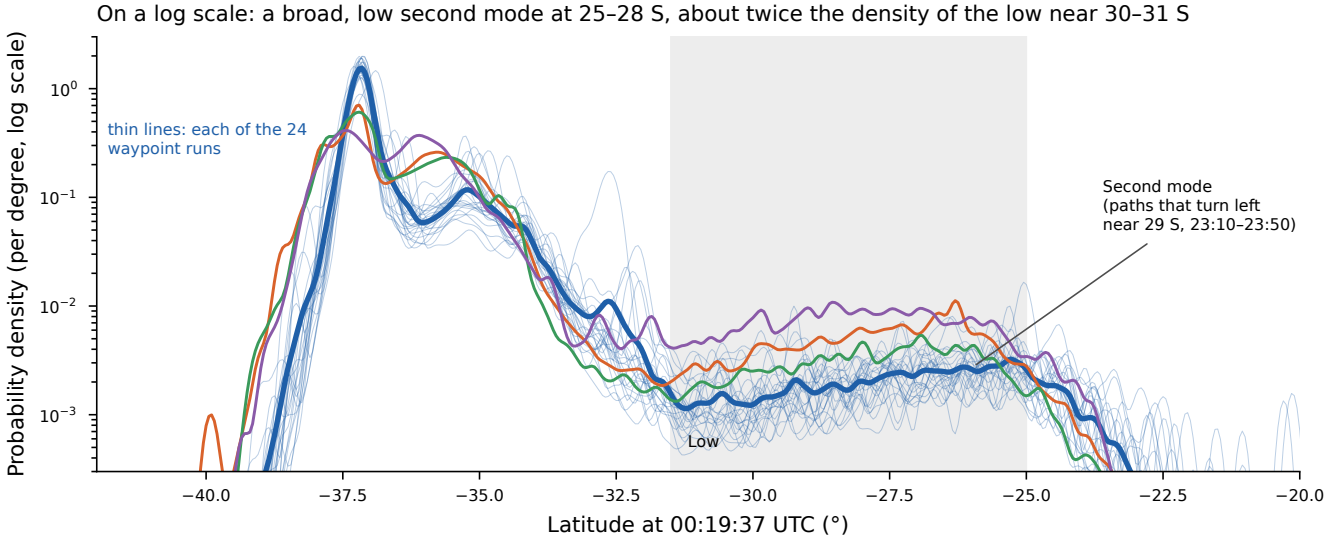
**a**




**b**




Plain (ASD-STE100): The curves show where the aircraft was at 00:19:37 UTC, by latitude, for each flight family of run J and before the drift and search evidence. Most probability is near 37° S. A small second peak is near 25–28° S, with a low near 30–31° S (grey band 31.5–25° S). Paths that end there make one large left turn near 29° S between about 23:10 and 23:50, after the 23:14 call attempt. This second peak holds 1.3% of probability in the waypoints family (which carries 99.8% of the result) and 2–5% in the other families. All 24 waypoint runs show it. Few independent paths carry it, so its exact shape is uncertain. The project lead accepts run J as converged (16 October 2026).

Technical: core run J (tag run-j = 1244b7c, deskstar x86_64), BTO+BFO case, weighted particle histograms with 0.1° Gaussian smoothing, latitude of the final state (00:19:37, 7th arc); families pooled over seeds (waypoints 24 × 7M, free 18 × 7M, descent-climb 16 × 3.5M, Davey dynamics 16 × 3.5M); posterior shares from ln Z at equal family prior. Mass 31.5–25° S: waypoints 1.3%, free 3.1%, descent-climb 2.0%, Davey 4.7%. Waypoints density 0.27%/deg at 26–25° S against 0.13%/deg at 31–30° S; 28–25° S exceeds 32–30° S in 24/24, 18/18, 16/16, 16/16 seeds. Pooled r-weighted effective lineages north of 33° S: waypoints about 120 (one lineage 8.4% of northern mass), free about 81. BFO sd 4.3177 Hz; 18:25:27 R600 held out; kappa_x N(1.0, 0.046) PROVISIONAL; one-engine s7 + s8 PROVISIONAL; 00:19 core treatment both BTO, no BFO. No drift, searched-area or end-of-flight evidence. Convergence: PASS by project ruling (16 Oct 2026). Built 18 Oct 2026.